Instruction for innstalling requirements

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1. Installation Steps Of Anaconda by Operating System

- **Windows**
  1. Download the https://www.anaconda.com/download.
  2. Double-click the .exe file to start the installation.
  3. Follow the on-screen instructions, accepting defaults (it is recommended to install for "Just Me").
  4. Verify the installation by opening the Anaconda Prompt from the Start Menu and running `conda list`.
- **Linux**
  1. Download the Linux installer (.sh file).
  2. Open the terminal, navigate to the folder with the download, and run: `bash <installer_name>.sh`.
  3. Follow the prompts, agree to the license terms, and confirm the installation location.
  4. Type "yes" to initialize Conda.
  5. Close and re-open the terminal to apply changes.
  6. Verify by running `conda list` in the terminal.
- **macOS**
  1. Download the appropriate installer for your Mac (Intel or M1/M2/M3).
  2. Run the installer package and follow the instructions.
  3. Verify by opening the terminal and running `conda list`.

Important Tips

- **Miniconda vs. Anaconda:** Miniconda is lightweight, containing only conda and Python. Anaconda includes GUI tools (Navigator) and many data science packages.
- **Path Configuration:** During Windows installation, you can choose to add Anaconda to your system PATH, although this is not always recommended as it may conflict with other applications.

- **Verification:** If conda list does not work, try conda --version to ensure it was added correctly.

2. Create a conda environment

- Open the Anaconda Prompt from the Start Menu of your laptop
- Type in the terminal: conda create -n nqs python=3.11.5
- When conda asks you to proceed proceed ([y]/n)?: Type y.
- After all installation, run: conda activate nqs, you will notice that (base) will change into (nqs).
- Download the requirements.txt file from my github page **https://github.com/Kenounouh/Machine-Learning-and-AI-for-Many-body-Physics-Lecture-Note-University-of-Dschang**
- Open the requirement.txt file and proceed to installation of all the 76 libraries
- To install a library run: pip install library_name. For example if you want to install absl-py==2.0.0, you run pip install absl-py==2.0.0.
- To install multiple libraries at the same time, run: pip install library_name1 library_name2 library_name3 … library_nameN. So, if I want to install absl-py==2.0.0, appnope==0.1.4, and asttokens==2.4.1, you need to run pip install absl-py==2.0.0 appnope==0.1.4 asttokens==2.4.1.
- You can add as many libraries as you want.

3. Activate the environment in VSCode

1. Installing VScode

- Download the https://code.visualstudio.com/docs?dv=win for Windows

- Once it is downloaded, run the installer (VSCodeUserSetup-{version}.exe)
  By default, VS Code is installed under C:\Users\{Username}\AppData\Local\Programs\Microsoft VS Code.

Tip
Setup adds Visual Studio Code to your %PATH% environment variable, to let you type 'code .' in the console to open VS Code on that folder. You need to restart your console after the installation for the change to the %PATH% environmental variable to take effect.

2. Activate environment

- Open VS Code

- Set up VS Code as you want

- Create a new file a choose Jupyter notbook file

- Select kernel

- Install the kernel from the VS Market
- Select a kernel again
- Choose python environment
- Choose nqs (python3.11.5)
- You are good to go !!!!